Tutorial Sheet-9

Indian Institute of Technology Jodhpur
MAL1010 (Mathematics-I)
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1. Find the limit points of the following sequencs:
   1. $(\frac{1}{n}, \frac{1}{n^2})$
   2. $(\frac{(-1)^n}{n^2}, \frac{n}{n+1})$
   3. $(\frac{(-1)^n}{n}, (1+\frac{1}{n})^n)$
2. Show that the sequence $\left\langle \left(mn^2\right) / \left(m^2+n^4\right)\right\rangle$ has its repeated limits but not the double limit.
3. Show that $\Sigma_{m,n=1}^{\infty} \frac{1}{(m+n)^{\alpha}}$ converges if $\alpha > 2$ and diverges if $\alpha < 2$.
4. Show that $\Sigma_{m,n=1}^{\infty} \frac{1}{m^{\alpha} n^{\beta}}$ converges if $\alpha > 1, \beta > 1$.
5. Give an example of a function $f : \mathbb{R}^2 \to \mathbb{R}$ such that
   - f is bounded below but not bounded above.
   - f is bounded above but not bounded below.
   - f is neither bounded below nor bounded above.
   - f is bounded.
6. Find the domain of the functions
   (a) $f(x,y) = \sqrt{y - x^2}$.
   (b) $f(x,y) = \frac{(x-1)(y+2)}{(y-x)(y-x^3)}$
   (c) $f(x,y) = cos^{-1}(y - x^2)$
7. Find the value of the limit of the following
   (a) $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$
   (b) $\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^2+y^2}$
   (c) $\lim_{(x,y)\to(0,0)} \frac{x^2-xy}{\sqrt{x}-\sqrt{y}}$
   Also, discuss the continuity at $(0,0)$ if the limit exists.
8. Consider the function $f(x,y)$ defined by:

$$f(x,y) = \begin{cases} x^2+y^2 & \text{if } x^2+y^2 \le 1, \\ 2-x^2-y^2 & \text{if } 1 < x^2+y^2 \le 2. \end{cases}$$

   Determine if $f(x,y)$ is continuous on the circle defined by $x^2+y^2 = 1$.

9. Examine the continuity of the function defined by:

$$f(x,y) = \begin{cases} x^2 + y^2 & \text{if } x^2 + y^2 < 1, \\ 1 & \text{if } x^2 + y^2 = 1, \\ 0 & \text{if } x^2 + y^2 > 1. \end{cases}$$

10. Prove that:

$$\lim_{(x,y)\to(2,3)} (x^2 + y^2) = 13$$

using the $\epsilon - \delta$ definition.